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Certification Dev IA

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Agenda

- Self introduction & objectives
- Educational projects presentation
  - ML : Are U a Legendary Pokemon ?
    - Corollary : what type of Pokemon are U ?
    - https://github.com/OrenChourakio6/projet-legendary-pokemon
  - DL : FingerPrint Validator
    - https://github.com/OrenChourakio6/projet-fingerprint-validator
- Summary & Course take-aways

Self Introduction & Objectives

- Software Developer
  - Master in Computer Engineering from ENSIMAG 1993
- +30 years experience in Electronic Design Automation
  - Last role as Senior Director managing R&D teams internationally
- Left position in Q3 2025 and in the process of relocating abroad in Q2 2026
- Desire to train and obtain certification in ML/AI in the meantime
- Exercise acquired ML & DL learnings into simple, straightforward classification mini projects

ML Project - Legendary Pokemon Classifier

- Why?
- Leverage a Pokemon dataset to predict whether a pokemon is legendary given its defensive and offensive characteristics
  - Corollary, use the dataset to predict the type of a given pokemon
  - https://www.kaggle.com/datasets/rounakbanik/pokemon
  - 800 pokemons, among which 70 are legendary
- Objective is to use different ML models on selected dataset to
  - Maximize accuracy
  - Adjust models hyper-parameters

ML Project - Legendary Pokemon Classifier

- Explore different dataset cleanup techniques
- Explore different ML models to classify pokemons
  - Logistic Regression, RandomForest, GradientBoosting
- Leverage Grid Search technique to explore hyper-parameters combination and improve models characteristics (size) and accuracy
- Training Results:
  - Binary classification (legendary): +99% accuracy is suspiciously too high, possible means strong auto-correlation between features.
  - Multi-Class classification (type) : very high accuracy (>90%)

DL Project - FingerPrintValidator

- Why?
- Objective is to determine whether a fingerprint image is readable
- Leverage transfer learning and fine-tuning techniques from a pre-trained model to solve a classification problem
  - Identifying finger name (Thumb, Index, Middle, Ring, Little) from a given image
  - Then rely on classification confidence percentage to determine whether picture is readable

DL Project - FingerPrintValidator

- Use SOCOFing dataset from Kaggle
  - https://www.kaggle.com/datasets/mreccentric/socofing-full-modified/data
  - 55270 grey scale, (96, 103) size images
    - 6000 original images
    - 49270 moderately to highly augmented images (cut, blur, drop)

Unaltered Cut Blur Drop

- Full dataset has a near uniform distribution between the 5 classes

- FingerPrint Dataset Creation Strategies:
  - 18k images : 6k original + 12k moderately altered
  - For each class:
    - Randomly split 1.2k original images between 800 train & 400 validation/test images
    - For each original image in train dataset, add corresponding altered images in the dataset, i.e. simulate data augmentation
    - Split 400 images into 2 distinct datasets for validation vs. test specifically with 2/3 vs. 1/3 ratio
  - To sum up : 16k images in train, 1.5k in validation, 500 in test datasets
- Train/validation/test datasets all have a near uniform class distribution

- CNN Model Architecture Exploration:
  - Consider MobileNet model family since image input size is small
    - MobileNet, MobileNetV2, MobileNetV3Small | Large
    - EfficientNet or ResNet family led to transfer learning top-1/top-2 accuracy results within +/- 2% for far less efficient #params & model size characteristics
  - Architecture : {MobileNet} + hidden dense layers + classification
    - One single hidden layer (512, 256)
    - Two hidden layers with 2**2 geometric reduction (256 + 128, 256 + 64)
    - Three hidden layers with 2**2 geometric reduction (512 + 128 + 32)
  - Hidden dense layers activation fonctions
    - ReLu vs. Leaky_ReLu vs. Gelu vs. Gelu_Approx

- Transfer Learning Training Strategies:
  - First axis : increase number of labels to classify
  - Second axis : grow the original dataset with moderately altered images
  - Third axis : explore model architectures and related activation functions
- Fine Tuning Training Strategies:
  - 2/3rd vs. 1/3rd of frozen vs. unfrozen layers in pre-trained base model
  - 2 options:
    - 1/3 unfrozen first layers vs. 2/3 unfrozen last layers
    - All unfrozen last layers

- Training Results:
  - Using enhanced dataset with simulated augmented images
  - Using MobileNetV2 with 2 distinct 256 & 64 hidden dense layers
  - Using leaky_relu activation function for hidden dense layers

Frozen MobileNetV2 + Dense 256 + Dense 64 + Classifier					
Transfer Learning	**Top1-Accuracy**	**Top2-Accuracy**	Fine Tuning	**Top1-Accuracy**	**Top2-Accuracy**
2 classes	87.9	n/a	**2 classes**	93	n/a
3 classes	**82.0**	**96.6**	**3 classes**	**83.6**	**97.4**
	65.6	**89.5**	**4 classes**	**69.7**	**90.8**
5 classes	**63.0**	**84.8**	**5 classes**	**66.9**	**85.9**

Summary & Course Take-aways

- Summary
  - Although being strictly educational projects, these 2 activities allowed me to grasp in vivo the concepts and technologies taught during the course
  - Feels like I visited 2 extremes of AI/ML:
    - Pokemon: hardly a ‘prediction’ since all possible data were given
    - Fingerprint: difficult to create dataset to train a model able to ‘generalize’ a prediction
- Course Take-aways
  - Expectations to understand what’s under the ML/DL hood - in terms of technical details, are met.
  - Feels confident to position myself as IA project development manager